# .NET Core Interviews: Your Ultimate Top 50 Questions Checklist

# Basic OOP Concepts

- **What is Object-Oriented Programming (OOP)?**
- **What are the four main principles of OOP?**
- **Explain Encapsulation with an example.**
- **How does Inheritance work in OOP?**
- **What is Polymorphism? Provide an example.**
- **What is Abstraction? How is it implemented in C#?**
- **What is the difference between an Abstract Class and an Interface?**
- **Can a class implement multiple interfaces? How?**
- **What is the difference between Method Overloading and Method Overriding?**
- **What is the difference between Compile-time Polymorphism and Runtime Polymorphism?**

# Classes & Objects

- **What is a Class in OOP?**
- **What is an Object in OOP?**
- **Can a class be inherited from multiple classes in C#?**
- **What is a Constructor? How many types of constructors are there?**
- **What is a Destructor in C#?**
- **What is the difference between Static, Sealed, and Partial classes?**
- **What is the use of the "this" keyword in C#?**
- **What is the difference between Value Types and Reference Types in C#?**
- **What is the difference between "new" and "override" keywords?**
- **What is the purpose of the "base" keyword in C#?**

# Interfaces & Abstract Classes

- **Can an abstract class have a constructor?**
- **Can an interface have a constructor?**
- **What is the default access modifier of interface methods?**
- **How do you implement multiple interfaces in a class?**
- **What is the difference between an Interface and an Abstract Class?**
- **Can an abstract class be sealed? Why or why not?**
- **What happens if a class does not implement all interface methods?**
- **Can an interface inherit another interface?**

# Encapsulation & Access Modifiers

- **What are access modifiers in C#?**
- **What is the difference between private, protected, internal, and public access modifiers?**
- **What is the use of the "protected internal" and "private protected" access modifiers?**
- **How does encapsulation improve code maintainability?**
- **Can we achieve encapsulation without using access modifiers?**

# Polymorphism (Method Overloading & Overriding)

- **Can static methods be overridden in C#?**
- **What is the difference between Method Hiding and Method Overriding?**
- **How does C# determine which method to call at runtime in case of method overriding?**
- **Can a method be overloaded based on the return type?**

# Inheritance

- **Can a class inherit from multiple classes in C#?**
- **What happens if a derived class does not call the base class constructor?**
- **Can a constructor be private? If so, in what scenarios?**
- **What is the difference between an IS-A and a HAS-A relationship in OOP?**
- **What is a virtual method in C#?**

# OOP & .NET Core Specific Questions

- **How does OOP help in designing scalable applications in .NET Core?**
- **What is Dependency Injection, and how does it relate to OOP principles?**
- **What is the SOLID principle? Explain each principle briefly.**
- **How is OOP implemented in ASP.NET Core applications?**
- **What is an Aggregate Root in Domain-Driven Design (DDD)?**
- **What is the difference between a Factory Pattern and a Repository Pattern?**
- **How do OOP principles help in writing unit-testable code in .NET Core?**
- **Can you explain how C# implements the Singleton Pattern?**